

Silicon Photonic MZIs and Bragg Filters: Design, Layout and Simulation of the v11 Circuit Set

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Abstract-- A five-circuit silicon photonic layout combines a course Mach-Zehnder interferometer (MZI), two native 500 nm MZIs with 25 and 100 μm path imbalances, and Bragg filters with 150 and 300 periods. A separate 500 $\times$ 220 nm MODE study gives a group index of 4.245970 at 1550 nm; circuit-fringe estimates are 4.1903 and 4.1899, lower by 1.31% and 1.32%. The MZI free spectral ranges are 22.934 and 5.734 nm, a ratio of 4.000. At 1550 nm, N300 Bragg transmission is 29.480 dB below N150. All circuits fit within 605 $\times$ 410 μm , with zero recorded local basic and functional layout markers. The preliminary 600 nm study is retained separately. Results are simulated; MODE sweep anomalies, convergence limits and circuit-model differences are explicitly documented.

Index Terms-- Mach-Zehnder interferometer, silicon photonics, group index, free spectral range, Bragg grating, layout.

I. INTRODUCTION

Mach-Zehnder interferometers provide a direct way to relate wavelength-dependent phase to waveguide dispersion [1]. This project uses unequal-arm MZIs so that the group index can be extracted from the measured free spectral range (FSR). Multiple path imbalances verify the expected scaling instead of relying on a single spectrum.

Sections III-IV retain the preliminary 600 nm MODE and ideal-circuit study. Sections V-X present the 500 nm MODE study, v11 layout, circuit spectra and local checks. The two widths represent different designs; their group-index values are not interchangeable.

II. THEORY

For an ideal lossless MZI, the propagation constant and normalized transmission are

$$\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda. \quad (1)$$

$$T(\lambda) = 1/2 [1 + \cos(\beta(\lambda) \Delta L)]. \quad (2)$$

The group index and approximate FSR near λ_0 are

$$n_g = n_{\text{eff}} - \lambda \, dn_{\text{eff}}/d\lambda. \quad (3)$$

$$\text{FSR} \approx \lambda_0^2 / (n_g \Delta L). \quad (4)$$

Adjacent spectral maxima or minima therefore provide a direct estimate of group index.

III. PRELIMINARY 600 NM WAVEGUIDE STUDY

The preliminary silicon strip waveguide is 600 nm wide and 220 nm high. MODE [2] was configured for the fundamental quasi-TE solution. At 1550 nm, $n_{\text{eff}} = 2.571982$, $n_g = 4.104976$, the TE polarization fraction is 99%, and the ideal material model has zero simulated propagation loss.

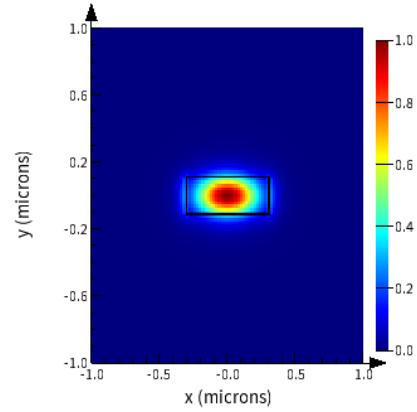


Fig. 1. Normalized electric-field intensity of the fundamental quasi-TE mode at 1550 nm.

The field is strongly confined in the silicon core. The exported 1500-1600 nm sweep contains 101 points. The real effective index decreases monotonically with wavelength and the imaginary part remains zero.

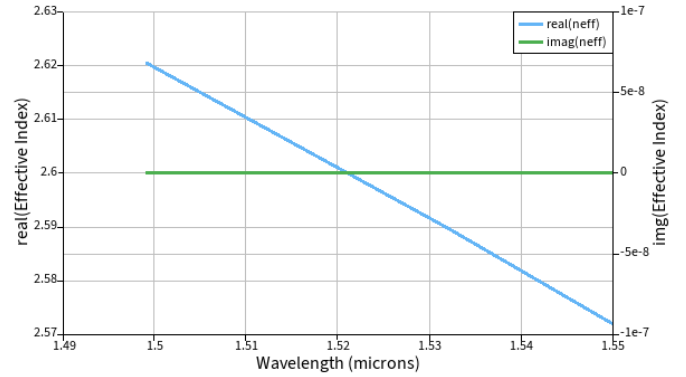


Fig. 2. Effective index versus wavelength. The displayed export covers 1500-1550 nm; the numerical data extend to 1600 nm.

The exported group-index data contain a discontinuity near 1532 nm. Because the tracked-mode overlap remains close to unity, the discontinuity is treated as a sweep/model-data artifact rather than a physical mode transition. The compact model is fitted only over the smooth 1540-1560 nm operating window.

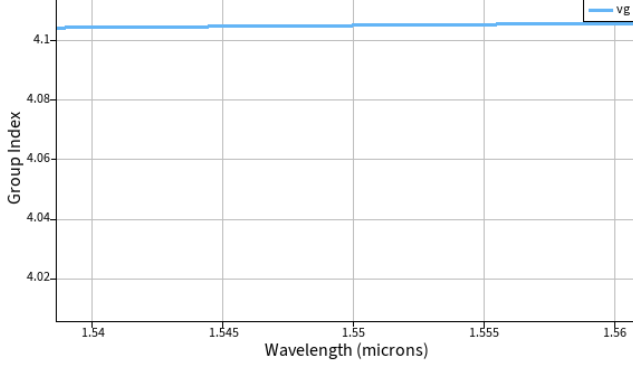


Fig. 3. Group index over the local compact-model window.

With $\Delta\lambda = \lambda - 1.55$ and wavelength in micrometres, the cubic fit is

$$n_{\text{eff}}(\lambda) = 2.5719816674 - 0.9890283430 \Delta\lambda - 0.0220137755 \Delta\lambda^2 + 0.0091898081 \Delta\lambda^3. \quad (5)$$

The maximum absolute residual over 1540-1560 nm is 9.3×10^{-11} . Differentiation gives $n_g(1550 \text{ nm}) = 4.104976$, consistent with the direct MODE result.

IV. PRELIMINARY IDEAL-MZI SIMULATION

Each INTERCONNECT circuit contains an ideal lossless 50:50 Y splitter, two MODE waveguide elements, and an ideal Y combiner. The upper arm is fixed at 100 microns and the lower arm ranges from 125 to 225 microns. The Optical Network Analyzer sweeps 1500-1550 nm with 2001 samples (25 pm spacing).

TABLE I
SIMULATED AND THEORETICAL FSR AT $\lambda_0 = 1525 \text{ nm}$

ΔL (μm)	FSR sim. (nm)	FSR theory (nm)	Group index extracted
25	23.070	22.662	4.032
50	11.436	11.331	4.067
75	7.652	7.554	4.052
100	5.730	5.665	4.059
125	4.582	4.532	4.061

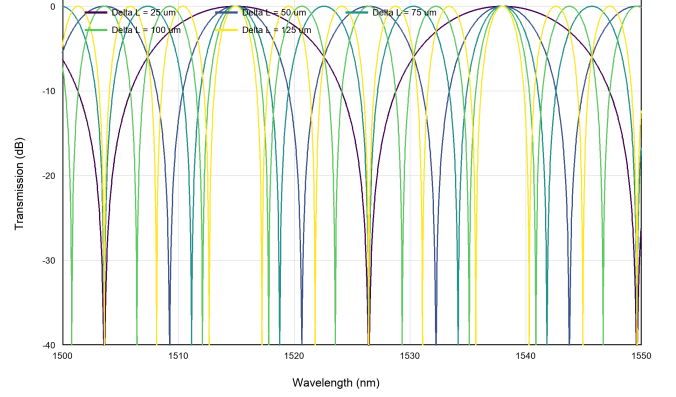


Fig. 4. Simulated transmission spectra. Increasing path imbalance increases the fringe density.

The spectra exhibit the expected increase in fringe density with ΔL , and the mean FSR agrees closely with theory.

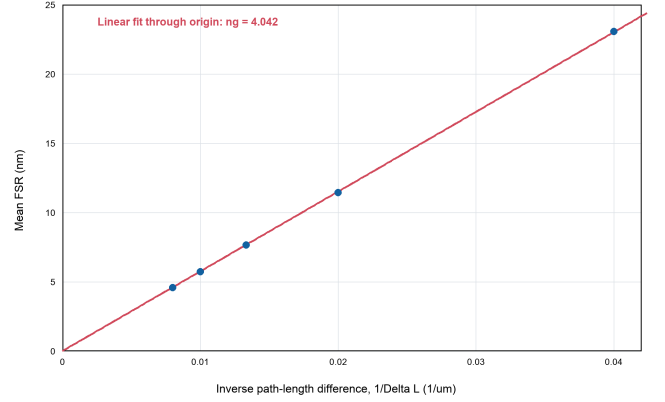


Fig. 5. Mean FSR versus $1/\Delta L$. The origin-constrained fit gives $n_g = 4.042$.

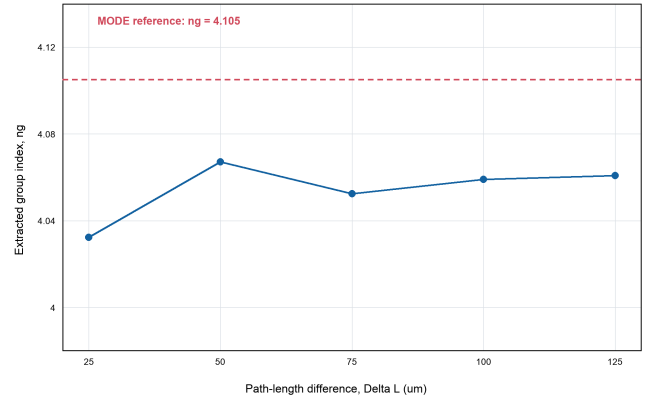


Fig. 6. Group index extracted independently from each MZI spectrum.

For the preliminary 600 nm design, the combined fit gives $n_g = 4.042$, which differs from MODE by -1.54%. The 25-micron design contains only about two complete periods and is less statistically robust; longer imbalances provide more extrema for averaging.

V. 500 NM MODE WAVEGUIDE STUDY

The separate MODE calculation uses a 500×220 nm rectangular silicon core with Si (Silicon) - Palik and a uniform, fixed-index background of 1.44401. The 2D Z-normal FDE solver is configured at 300 K in a 2×2 μm transverse domain, with 100×100 mesh cells and Metal boundaries on all four transverse sides. These settings are documented by supplied screenshots; the LMS project was not supplied.

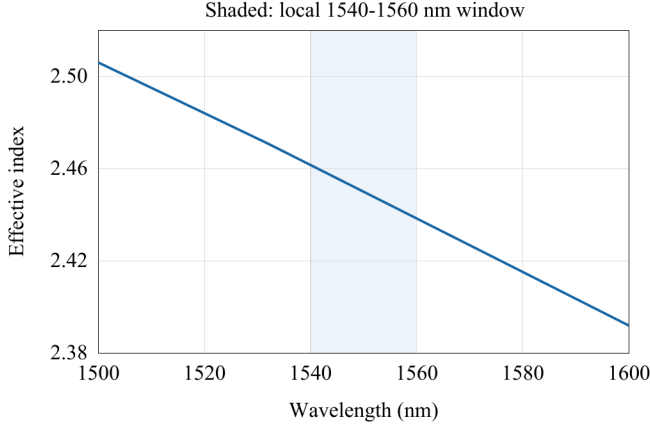


Fig. 7. Real effective index from the complete 101-point 500 nm MODE export. Shading marks the local fit window; no full-range samples are removed.

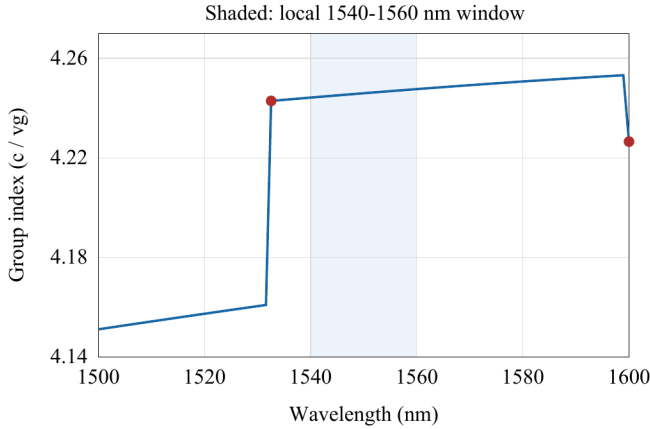


Fig. 8. Group index computed as c/v_g . Red markers flag the step near 1532.6 nm and the 1600 nm endpoint anomaly; the full exported curve is retained.

The frequency sweep covers 1500-1600 nm with 101 points, mode tracking and detailed dispersion enabled. Tracked-mode overlap lies between 0.9999962 and 0.9999997. The 1550 nm screenshot reports 98% TE polarization for the selected fundamental-like mode; exported imaginary effective index and loss are zero.

At 1550 nm, linear interpolation of the numerical export gives $n_{\text{eff}} = 2.450006$ and $n_g = 4.245970$. A local cubic fit to n_{eff} over 1540-1560 nm (20 sampled points) gives $n_g = 4.245970$ from (3), agreeing with c/v_g to better than 10^{-6} . The maximum effective-index fit residual is 2.49×10^{-10} ; this measures local fit consistency, not physical simulation accuracy.

TABLE II
GROUP-INDEX COMPARISON AT 1550 NM

Method / model	n_g	Difference*
500 nm MODE, c/v_g	4.245970	Reference
500 nm MODE, local fit	4.245970	<0.0001%
A25 native circuit	4.190280	-1.3116%
A100 native circuit	4.189874	-1.3212%

* $100 \times (n_{g,\text{circuit}} / n_{g,\text{MODE}} - 1)$, with the same MODE reference used for both circuits.

The exported group index rises from 4.160908 at 1531.589 nm to 4.242887 at 1532.567 nm. At the final sample it decreases from 4.253211 at 1598.934 nm to 4.226535 at 1600 nm. Neither feature lies within the local 1540-1560 nm fit. Near-unity overlap supports mode continuity but does not establish the cause of these anomalies. Their origin remains unresolved.

The local MODE and circuit estimates differ by about 1.3%. The MODE background is nondispersive, while equivalence to the EBeam compact-model material stack and dispersion has not been demonstrated. Mesh/domain convergence has not been tested. Accordingly, the difference is reported as a model comparison rather than attributed to a specific physical mechanism or treated as validated fabrication accuracy.

The full export, overlap values, screenshot settings, fit coefficients and source hashes are preserved with the report. Future convergence checks should retain the original data and compare the same tracked mode. The existing v11 native compact models and circuit simulations are unchanged.

VI. V11 LAYOUT IMPLEMENTATION

Five-circuit silicon layout — v11

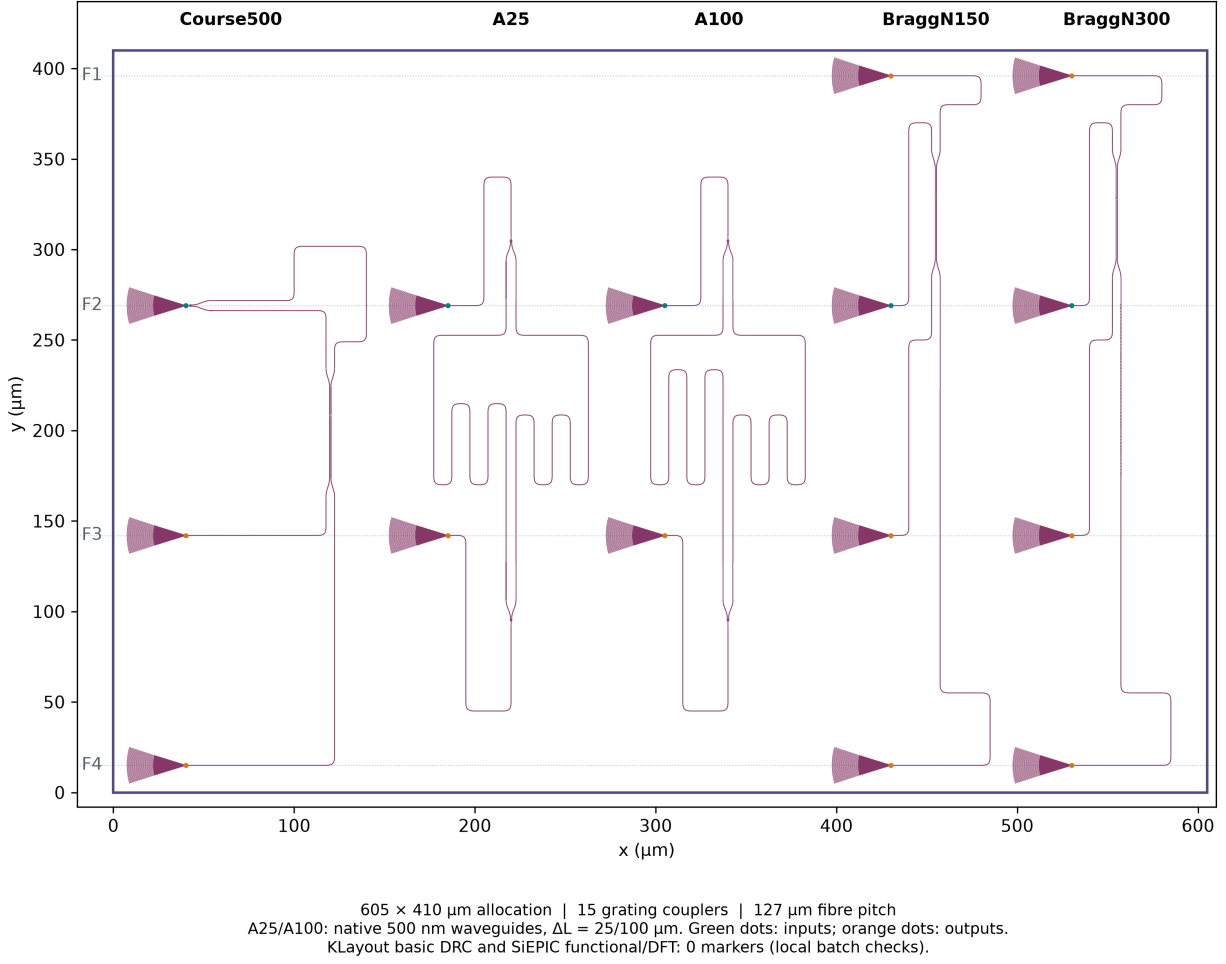


Fig. 9. Silicon geometry of EBeam_Macwu_v11. The five independently addressable circuits occupy a $605 \times 410 \mu\text{m}$ allocation. Green and orange markers denote input and output coupler pins, respectively.

The v11 layout uses native SiEPIC EBeam cells [3]. Course500 retains the course Y-splitter/BDC-combiner MZI [4]. A25 and A100 use Y splitters and combiners with 500 nm waveguides; their routed arm imbalances are 25.000 and 100.000 μm . Four former taper sections per circuit are now 20 μm straight 500 nm segments. The 600 nm preliminary model is not used for these circuits.

BraggN150 and BraggN300 contain 150 and 300 periods, respectively. The period is 317 nm, the corrugation width is 50 nm, and the nominal waveguide width is 500 nm. The profile is rectangular with zero sidewall misalignment. Grating lengths are 47.55 and 95.10 μm .

The 15 grating couplers share a 127 μm vertical fibre pitch and R0 orientation. The common input row is F2 ($y = 269 \mu\text{m}$). A25/A100 use F3 for output; Course500 uses F3/F4. Each Bragg circuit uses F1 for reflection, F3 for reference and F4 for transmission.

GC pin columns are $x = 40, 185, 305, 430$ and $530 \mu\text{m}$. Recorded fibre targets lie 20.4 μm to the left of each pin column. All connecting nets are 500 nm wide, with a 5 μm TE bend setting. Course500 and the Bragg structures retain their verified v10 silicon, pins and device-recognition geometry.

VII. V11 MZI SIMULATION RESULTS

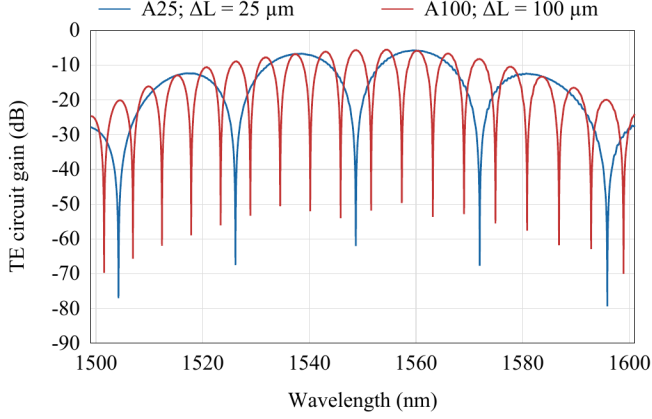


Fig. 10. User-exported A25/A100 ONA input 1, mode 1 TE circuit gains. Each spectrum contains 5,000 samples over 1499.019-1601.000 nm.

TABLE III
NATIVE 500 NM MZI RESULTS

Quantity	A25	A100
Routed arms (μm)*	363.975 / 388.975	363.975 / 463.975
ΔL (μm)	25.000	100.000
FSR at 1550 nm (nm)	22.934	5.734
Inferred n_g at 1550 nm	4.1903	4.1899
Minima / intervals	5 / 4	18 / 17

*Excludes two matching 20 μm straight sections per branch. Including them preserves ΔL.

Gain and wavelength rows are paired before sorting. Strict local sampled minima below -35 dB define adjacent intervals. No smoothing or notch-depth interpolation is applied. For wavelengths and lengths in consistent units, the interval index [5] is

$$n_{g,int} = \lambda_1 \lambda_2 / [\Delta L (\lambda_2 - \lambda_1)]. \quad (6)$$

Interval indices are linearly interpolated between midpoint wavelengths to 1550 nm; (4) then gives the equivalent local FSR. The ratio is 3.9996, consistent with inverse-ΔL scaling. These native-circuit indices are compared with the separate MODE result in Table II; material and model equivalence remains unestablished.

A100 retains the wavelength header a25_wavelength_nm. Its numeric grid exactly matches A25; pairing assumes identical sweep settings. Circuit identity follows supplied filenames and conversation because ICP projects were not provided.

VIII. BRAGG PERIOD-COUNT STUDY

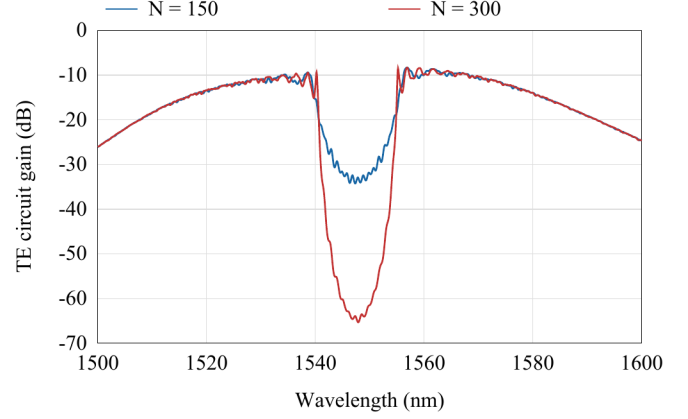


Fig. 11. User-exported Bragg ONA input 3, mode 1 transmission gains: 2,000 samples over 1500.047-1600.000 nm.

TABLE IV
BRAGG CIRCUIT TRANSMISSION

Quantity	N150	N300
Grating length (μm)	47.55	95.10
Sampled minimum (dB)	-34.242	-65.310
Minimum wavelength (nm)	1547.339	1547.888
Gain at 1550 nm (dB)	-31.996	-61.477

These numerical exports originate from v10. The v11 verification records unchanged Bragg silicon, pins and DevRec, supporting reuse with explicit provenance. The broadband directional coupler provides separate reflection, reference and transmission ports.

At 1550 nm, linear interpolation in dB gives N300 a 29.480 dB lower transmission than N150. This is a full-circuit result including grating couplers, directional coupler and routing, not intrinsic grating rejection. Different sampled-minimum wavelengths do not establish a physical shift of the Bragg centre.

Numeric reference-channel spectra were not supplied. Reference-normalized suppression and bandwidth therefore remain unreported. The native Bragg compact model also does not independently characterize the 10 nm terminal landing used to meet the declared 500 nm port width.

IX. COURSE BASELINE AND EVIDENCE SCOPE

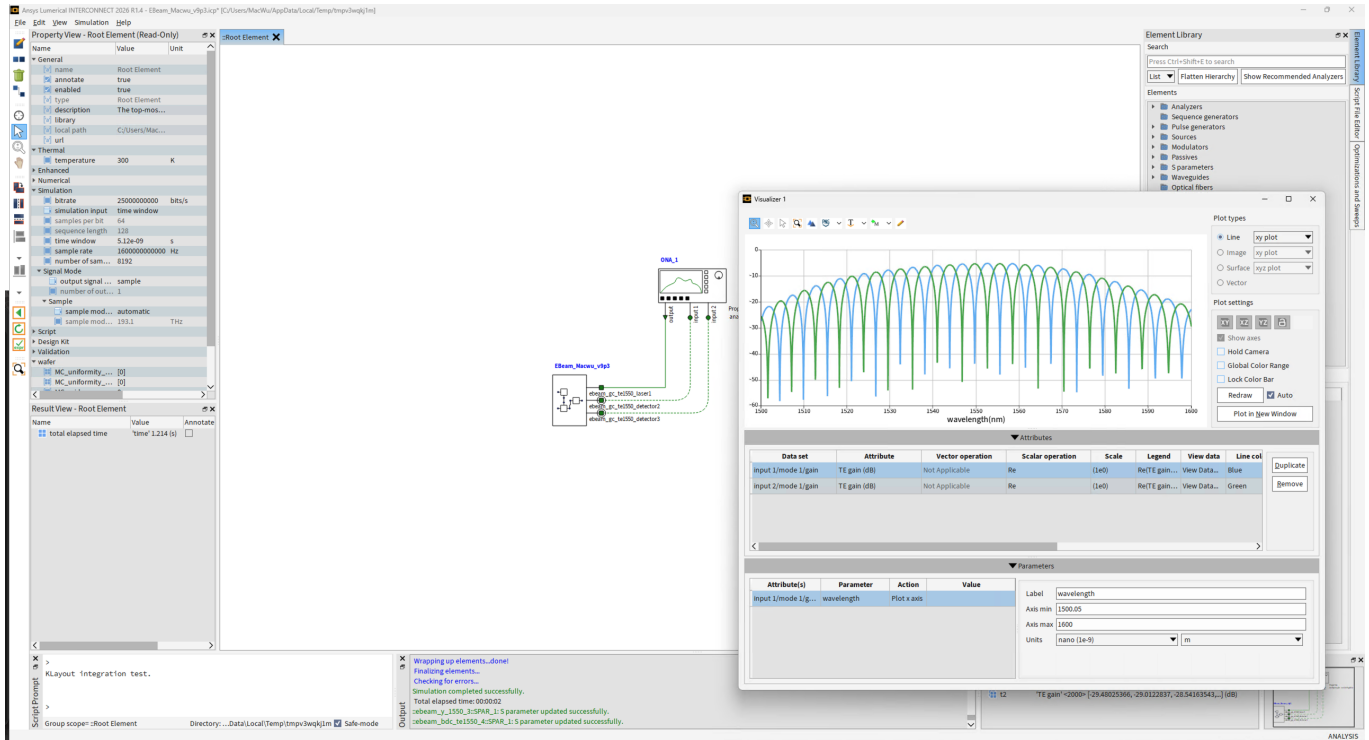


Fig. 12. Course500 screenshot supplied on 4 October 2026. The displayed project is EBeam_Macwu_v9p3; the log shows successful completion and two alternating output spectra. It is earlier qualitative evidence, not an authenticated v11 ICP run. No numeric FSR is inferred from the screenshot.

Course500 preserves the supplied course exercise: a TE grating input, Y branch, unequal 500 nm arms, broadband directional-coupler combiner and two grating outputs. Its routed-arm imbalance is 99.293 μm . The supplied screenshot supports qualitative two-output behaviour; paired numerical exports remain unavailable.

Recorded local checks use KLayout 0.30.12, SiEPIC-Tools 0.5.31 and EBeam PDK 0.4.53. The verification record contains 136 deterministic checks, with zero basic PDK DRC markers and zero functional/DFT markers. The optical graph contains 55 components, 53 nets and 106 connected optical pins. Five unique opt_in labels and isolated netlists preserve separate circuit access.

Build-time runtime status is retained in the original verification record. The newer A25/A100 exports are documented separately in the 5 October evidence set. These data are simulations, not fabricated-device measurements; local checks do not constitute full foundry signoff or full course-rubric certification.

X. DISCUSSION AND CONCLUSION

The 500 nm MODE study gives $n_g = 4.245970$ at 1550 nm; native-circuit estimates are lower by 1.31-1.32%. This is a cross-model comparison, not a converged PDK validation. A25/A100 retain approximately 4:1 FSR scaling, and doubling the Bragg period count lowers transmission near 1550 nm. The 600 nm results remain a separate preliminary study.

Future measurements should excite F2 in TE and record the designated output rows at each target column. Acquire Bragg reflection, reference and transmission together before normalized bandwidth analysis. MODE mesh/domain convergence, ICP identity, fabrication variability and the Bragg terminal-landing model remain coverage gaps. Original numerical exports, layout hashes and verification records accompany the report provenance.

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.
- [2] Ansys, "Lumerical MODE and INTERCONNECT documentation," accessed Sep. 2026.
- [3] SiEPIC, *SiEPIC EBeam PDK*, installed v0.4.53. github.com/SiEPIC/SiEPIC_EBeam_PDK.
- [4] UBCx Phot1x, *Layout of an MZI*, supplied course materials, 2026.
- [5] Ansys Optics, "Optical filters," FSR relation; cited in the 5 Oct. 2026 simulation evidence report.